

Non-classical Modal Logics

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Available at www.leonardopacheco.xyz/slides/aizu.pdf

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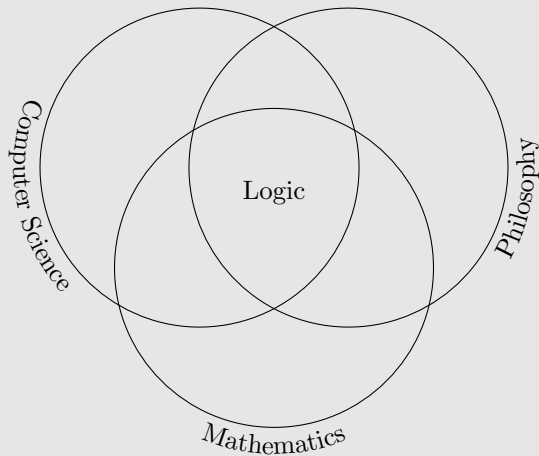
CAREER OVERVIEW — EDUCATION

- ▶ 2012–2018: Universidade Federal de Goiás, Brazil
 - ▶ Two years as undergraduate in Computer Science
 - ▶ One year as Exchange Student at Tohoku University
 - ▶ Undergraduate in Mathematics
- ▶ 2018–2023: Tohoku University, Japan
 - ▶ M.Sc. in Mathematics, supervised by Kazuyuki Tanaka
 - ▶ Ph.D. in Mathematics, supervised by Keita Yokoyama
 - ▶ TA for Mathematical Logic and Computability courses
 - ▶ Coordinated seminars for graduate students in English
 - ▶ Kawai Doctoral Thesis Award (Tohoku University)

CAREER OVERVIEW — RESEARCH

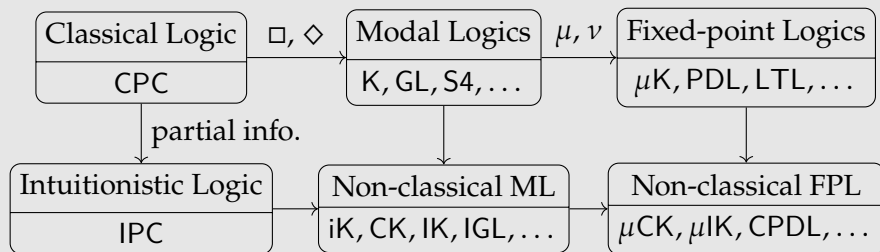
- ▶ 2023–2024: TU Wien, Austria
 - ▶ Postdoctoral researcher
 - ▶ Supervised by Juan P. Aguilera
- ▶ 2024–present: Institute of Science Tokyo, Japan
 - ▶ JSPS Postdoctoral Fellow
 - ▶ KAKENHI Grant 24KF0214:
“Constructive μ -calculus for model checking”
 - ▶ Supervised by Ryo Kashima
- ▶ Reviewing:
 - ▶ Fuzzy Sets and Systems, Logic Journal of the IGPL, Annals of Pure and Applied Logic, IJCAR 2026, AiML 2026, LICS 2026, TABLEAUX 2025
 - ▶ Program Committee of 8th Intuitionistic Modal Logic and Applications Workshop (July 2026, Lisbon)

LOGIC IS HERE:



BACKGROUND

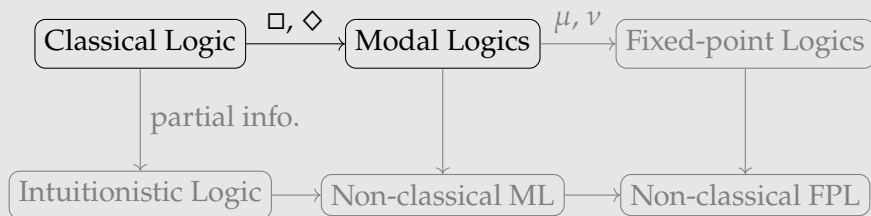
I work with the combination of modal logics, intuitionistic logic, and fixed-point logics.



SOME HISTORY

- ▶ Modal Logics
 - ▶ Modern semantics (Kripke, 1959)
 - ▶ Reasoning about programs (Fischer and Ladner, 1977)
- ▶ Fixed-point Logics:
 - ▶ μ -calculus (Kozen, 1983)
 - ▶ Axiomatization (Walikiewicz, 2000)
- ▶ Intuitionistic Logic
 - ▶ Intuitionism (Brouwer, 1923)
 - ▶ A logic of partial information (Vickers, 1989)
- ▶ Non-classical Modal Logic
 - ▶ Some scattered old works: (Ono, 1977) and (Simpson, 1994)
 - ▶ Surge of research since (Das and Marin, 2023)

MODAL LOGICS



MODAL LOGICS

Modal logics add to propositional logic:

- ▶ $\Box$: necessity;
- ▶ $\Diamond$: possibility.

These allow us to state *how* statements are true.

Example

Three interpretations of $\Box\varphi$:

- ▶ *φ is provable;*
- ▶ *φ holds after the execution of the program p ;*
- ▶ *the agent a knows that φ .*

MODELS FOR MODAL LOGIC

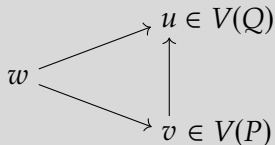
A model for modal logic is:

$$M = \langle W, R, V \rangle,$$

where:

- ▶ $\langle W, R \rangle$ is a directed graph;
- ▶ $V : \text{Prop} \rightarrow \mathcal{P}(W)$ specifies the states where each proposition is true.

Example

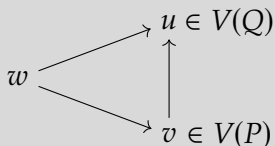


INTERPRETING FORMULAS

Let $M = \langle W, R, V \rangle$ and $w \in W$, then

- ▶ $M, w \models \Box\varphi$ if, for all v , if wRv then $M, v \models \varphi$;
- ▶ $M, w \models \Diamond\varphi$ if there is v such that wRv and $M, v \models \varphi$.

Example



- ▶ $M, w \models \Box(P \vee Q)$
- ▶ $M, v \not\models \Diamond P$

THE MODAL LOGIC K

We define the modal logic K as the set of formulas valid over all models:

Definition

Let φ be a formula.

Then $K \models \varphi$ if, for every model $M = \langle W, R, V \rangle$ and $w \in W$,

$$M, w \models \varphi.$$

PROOF SYSTEMS

We can also define logics via proof systems.

Example

A proof of K is a finite labeled tree:

$$\begin{array}{c}
 \frac{\varphi \vdash \varphi, \psi \quad \varphi, \psi \vdash \psi}{\varphi \rightarrow \psi, \varphi \vdash \psi} \\
 \frac{\square(\varphi \rightarrow \psi), \square\varphi \vdash \square\varphi}{\square(\varphi \rightarrow \psi) \vdash \square\varphi \rightarrow \square\psi} \\
 \hline
 \vdash \square(\varphi \rightarrow \psi) \rightarrow (\square\varphi \rightarrow \square\psi)
 \end{array}$$

The semantics and proof-theoretic definitions of K coincide:

Theorem (Soundness and Completeness)

For every formula φ , $K \models \varphi$ if and only if $K \vdash \varphi$.

OTHER MODAL LOGICS

We can obtain other logics by modifying our semantics or our proofs:

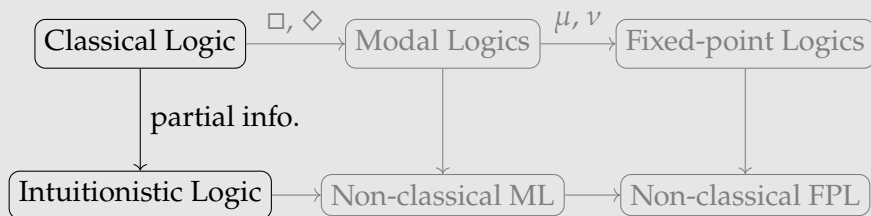
Example

K4 is the logic obtained by

- ▶ *requiring R to be transitive; or*
- ▶ *adding the rule*

$$\frac{\Gamma, \Box\Gamma \vdash \varphi}{\Box\Gamma \vdash \Box\varphi}$$

INTUITIONISTIC PROPOSITIONAL LOGIC



INTUITIONISTIC PROPOSITIONAL LOGIC

In intuitionistic logic, we cannot use:

- ▶ excluded middle: $\varphi \vee \neg\varphi$,
- ▶ double negation exclusion: $\neg\neg\varphi \rightarrow \varphi$.

More precisely, proof carries evidence:

- ▶ To prove $\varphi \vee \psi$, we prove one of φ or ψ and identify which one;
- ▶ To prove $\varphi \rightarrow \psi$, we define a construction that converts proofs of φ into proofs of ψ .

INTUITIONISTIC LOGIC AND PARTIAL INFORMATION

A model for intuitionistic propositional logic is:

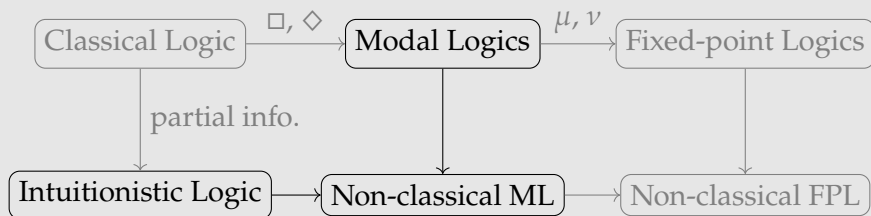
$$M = \langle W, \leq, V \rangle,$$

where:

- ▶ $\langle W, \leq \rangle$ is a directed graph;
- ▶ $\leq$ is a reflexive and transitive relation;
- ▶ $V : \text{Prop} \rightarrow \mathcal{P}(W)$ satisfies

$$w \leq v \text{ and } w \in V(P) \text{ imply } v \in V(P).$$

INTUITIONISTIC PROPOSITIONAL LOGIC



NON-CLASSICAL MODAL LOGIC

A model for the non-classical modal logic CK is

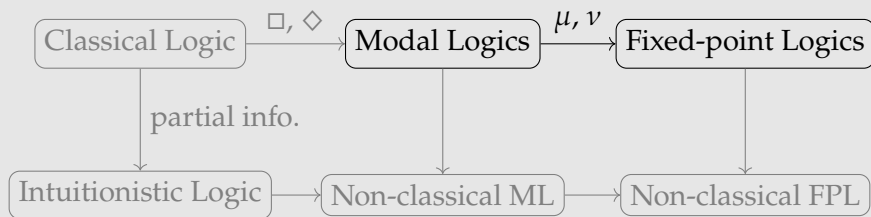
$$M = \langle W, \leq, R, V \rangle,$$

where:

- ▶ $\langle W, R, V \rangle$ is a model for modal logic, and
- ▶ $\langle W, \leq, V \rangle$ is a model for intuitionistic propositional logic.

There are choices for the interpretations of $\Box$ s and $\Diamond$ s, we do not discuss these in this presentation.

FIXED-POINT LOGICS



FIXED-POINT LOGICS

We can express “infinite” properties by finite formulas if we use fixed-points:

Example

Interpret $\Box\varphi$ as “ φ holds after the execution of a program p ”, then

$$\nu X.(\varphi \wedge \Box X) \approx \varphi \wedge \Box\varphi \wedge \Box\Box\varphi \wedge \dots,$$

that is,

$$\nu X.(\varphi \wedge \Box X) \approx “\varphi \text{ always holds}”.$$

The μ -calculus is a modal logic containing both least and greatest fixed-points. It balances expressivity and computational complexity:

- ▶ validity is decidable in EXPTIME-complete,
- ▶ model checking is polynomial time, in general.

FIXED-POINTS, SEMANTICALLY

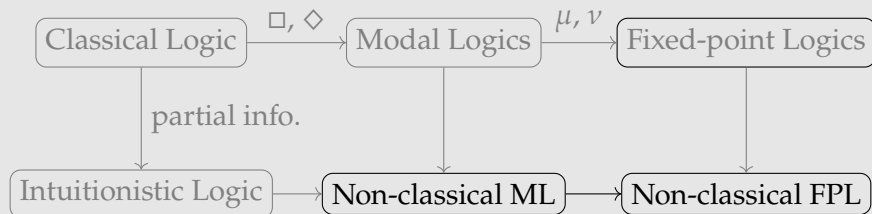
Let M be a model, then $\nu X.(\varphi \wedge \Box X)$ is the greatest fixed-point of the operator

$$\Gamma : \mathcal{P}(W) \rightarrow \mathcal{P}(W)$$

$$A \mapsto \{w \mid M[X \mapsto A], w \models \varphi \wedge \Box X\},$$

where $M[X \mapsto A]$ is the same as M , except $V(X) = A$.

MY WORK



Objective

Better logics to reason about programs with partial information.

NON-WELLFOUNDED PROOF SYSTEMS

Another way to get new logics is to allow infinite proofs:

$$\begin{array}{c}
 \vdots \\
 \hline
 \frac{\Box(\Box P \rightarrow P) \vdash P, \Box P \quad \Box(\Box P \rightarrow P), P \vdash P}{\Box(\Box P \rightarrow P), \Box P \rightarrow P \vdash P} \\
 \hline
 \frac{\Box(\Box P \rightarrow P) \vdash P, \Box P \quad \Box(\Box P \rightarrow P), P \vdash P}{\Box(\Box P \rightarrow P), \Box P \rightarrow P \vdash P} \\
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 \frac{\Box(\Box P \rightarrow P) \vdash P, \Box P \quad \Box(\Box P \rightarrow P), P \vdash P}{\Box(\Box P \rightarrow P), \Box P \rightarrow P \vdash P} \\
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 \frac{\Box(\Box P \rightarrow P) \vdash P, \Box P \quad \Box(\Box P \rightarrow P), P \vdash P}{\Box(\Box P \rightarrow P), \Box P \rightarrow P \vdash P} \\
 \hline
 \frac{\Box(\Box P \rightarrow P) \vdash P, \Box P \quad \Box(\Box P \rightarrow P), P \vdash P}{\Box(\Box P \rightarrow P), \Box P \rightarrow P \vdash P} \\
 \hline
 \vdash \Box(\Box P \rightarrow P) \rightarrow \Box P
 \end{array}$$

GL proves $\Box(\Box P \rightarrow P) \rightarrow \Box P$.

CYCLIC PROOF SYSTEMS

Better yet, use cyclic proofs to encode infinite proofs:

$$\begin{array}{c}
 \hline
 \textcolor{red}{\Box(\Box P \rightarrow P) \vdash P, \Box P} \quad \textcolor{red}{\Box(\Box P \rightarrow P), P \vdash P} \\
 \hline
 \textcolor{red}{\Box(\Box P \rightarrow P), \Box P \rightarrow P \vdash P} \\
 \hline
 \begin{array}{cc}
 \textcolor{blue}{\Box(\Box P \rightarrow P) \vdash P, \Box P} & \textcolor{blue}{\Box(\Box P \rightarrow P), P \vdash P} \\
 \hline
 \textcolor{blue}{\Box(\Box P \rightarrow P), \Box P \rightarrow P \vdash P} \\
 \hline
 \textcolor{blue}{\Box(\Box P \rightarrow P) \vdash \Box P} \\
 \hline
 \vdash \Box(\Box P \rightarrow P) \rightarrow \Box P
 \end{array}
 \end{array}$$

GL proves $\Box(\Box P \rightarrow P) \rightarrow \Box P$.

INTUITIONISTIC GÖDEL–LÖB IGL

Theorem (Das, van der Giessen and Marin, 2024)

Assume Σ_1^1 -determinacy. The non-wellfounded proof system $m\ell\text{IGL}$ is sound and complete with respect to IGL-models.

Note: Σ_1^1 -determinacy is not provable over ZFC.

Theorem (Aguilera and Pacheco, 2025)

The cyclic proof-system $cml\text{IGL}$ is sound and complete with respect to IGL-models.

- ▶ No strong assumption.
- ▶ IGL is semi-decidable.
- ▶ First work combining labeled and cyclic proof systems.

CONSTRUCTIVE μ -CALCULUS

We may evaluate $M, w \models \varphi$ by a game between two players:
Verifier and Refuter.

Theorem (Pacheco, 2026, preprint)

- ▶ *Relational semantics and game semantics for the constructive μ -calculus are equivalent.*
- ▶ *The non-wellfounded proof system $\text{lab}\mu\text{CK}_\omega$ is sound and complete with respect to the constructive μ -calculus.*
- ▶ First work on the full μ -calculus in a non-classical setting.
- ▶ Extends to other non-classical μ -calculi.
- ▶ Non-wellfounded proofs as universal strategies for valid formulas.

TOPOLOGICAL SEMANTICS

Theorem (McKinsey and Tarski, 1944)

S4 is complete with respect to topological spaces if we interpret $\Box$ as the interior operator.

Theorem (Montacute, Pacheco, submitted)

iK4, iGL, iS4, iGrz, iCK4, IEL, PLL, mHC, iSL, and KM are complete with respect to their topological models.

Theorem (Aguilera, Fernández-Duque, Pacheco, preprint)

CK4, IK4, GK4, CS4, IS4, and GS4 are complete with respect to their topological models.

TOPOLOGIES, $\square$ S AND $\diamond$ S

- ▶ iK4 has only $\square$ s.
- ▶ iK4-models need only one topology:

$$\langle W, \tau_{\square}, V \rangle.$$

- ▶ IK4 has both $\square$ s and $\diamond$ s.
- ▶ IK4-models need three topologies:

$$\langle W, \tau_{\rightarrow}, \tau_{\square}, \tau_{\diamond}, V \rangle.$$

INTUITIONISTIC EPISTEMIC LOGIC

The logic IEL formalizes:

intuitionistic truth implies intuitionistic knowledge.

Theorem (Pacheco, Sedlár, in preparation)

The following two views are compatible with IEL:

- ▶ *knowledge is necessity ($\Box$);*
- ▶ *knowledge is possibility ($\Diamond$).*

OTHER RESULTS

μ -calculus' expressivity over various classes of models:

- ▶ Pacheco, "The μ -calculus' Alternation Hierarchy is Strict over Non-Trivial Fusion Logics", 2025.
- ▶ Pacheco, Tanaka, "The alternation hierarchy of the mu-calculus over weakly transitive frames", 2022.

Connection between computation, fixed-points, large ordinals, and infinite games:

- ▶ Aguilera, Lubarsky, Pacheco, "Higher-Order Feedback Computation", 2024.

Reverse Mathematics of infinite games and reflection principles:

- ▶ Pacheco, Yokoyama, "Determinacy and reflection principles in second-order arithmetic", arXiv:2209.04082.

RESEARCH PLANS

- ▶ Short-term: basic theory of non-classical logics.
 - ▶ Decidability of IGL.
 - ▶ Cyclic proof system and decidability of non-classical μ -calculi.
- ▶ Mid-term: application of the basic theory.
 - ▶ Model checking: Does a finite-state model satisfy a given specification?
 - ▶ Knowledge representation: Represent information in a structured, machine-processable form.
- ▶ Long-term: other non-classical logics.
 - ▶ Paraconsistent logic: allows contradictory information.

FUTURE PROSPECTS

- ▶ Better logics to reason about programs with partial information.
- ▶ Along the way, solve some of the many open problems about non-classical modal logics.

Thank you!

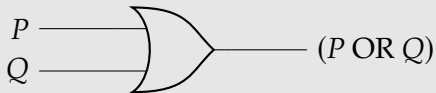
模擬講義 命題論理の意味論

カワカミパシェコ レオナルド
東京科学大学・情報理工学院

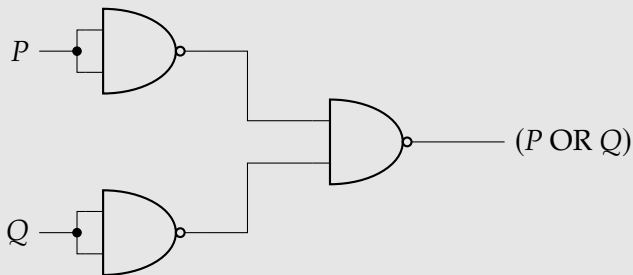
2026 年 8 月 4 日

完全性と論理回路

論理回路 OR を以下で表す：



OR は NAND だけで表せる：



講義内容

- ▶ 命題とは
- ▶ 意味論：割り当てと真理値表
- ▶ 同値
- ▶ 演算子の完全性

命題とは

真 (T) か偽 (F) かが定まる主張を命題 (proposition) と呼ぶ。

例

- ▶ 「8月4日に会津若松市で雪が降ります」は偽である。
- ▶ 「任意の実数 x に対し、 $\sqrt{x}$ は実数である」は偽である。
- ▶ 「 $2 + 2 == 4$ 」は真である。

命題を形式的に表すため、命題記号の集合

$$\text{Prop} = \{P, Q, R, S, \dots\}$$

を固定する。

論理式

演算子 $\neg$ 、 $\wedge$ 、 $\vee$ 、 $\rightarrow$ を使って、論理式を定義する：

定義

- ▶ 命題記号は論理式である。
- ▶ φ と ψ が論理式ならば、

$$(\neg\varphi), (\varphi \wedge \psi), (\varphi \vee \psi), (\varphi \rightarrow \psi)$$

はいずれも論理式である。

以上に該当する記号列以外は論理式ではない。

以下のスライドでは、最外側の括弧を省略する。

論理式の例

例

- ▶ $(\neg P) \rightarrow Q$ は論理式である。
- ▶ $(\rightarrow P) \rightarrow Q$ は論理式でない。

真理値割り当て

定義

$V : \text{Prop} \rightarrow \{T, F\}$ は命題記号に対する真理値割り当て (*valuation*) である。

例

$V(P) = T, V(Q) = F, V(R) = F, \dots$

命題記号に対する割り当て V が与えられたときに、 V を任意の論理式に拡張することができる。

「でない」の真理値表

V を任意の論理式に拡張するために、以下のような真理値表 (truth table) を使う：

P	$\neg P$
T	
F	

「でない」の真理値表

V を任意の論理式に拡張するために、以下のような真理値表 (truth table) を使う：

P	$\neg P$
T	F
F	T

例

$V(P) = T$ ならば、 $V(\neg P) = F$ である。

「かつ」の真理値表

P	Q	$P \wedge Q$
T	T	
T	F	
F	T	
F	F	

「かつ」の真理値表

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

例

$V(P) = F$ かつ $V(Q) = T$ ならば、 $V(P \wedge Q) = F$ である。

「または」の真理値表

P	Q	$P \vee Q$
T	T	
T	F	
F	T	
F	F	

「または」の真理値表

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

例

$V(P) = F$ かつ $V(Q) = T$ ならば、 $V(P \vee Q) = T$ である。

「ならば」の真理値表

P	Q	$P \rightarrow Q$
T	T	
T	F	
F	T	
F	F	

「ならば」の真理値表

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

例

$V(P) = T$ かつ $V(Q) = F$ ならば、 $V(P \rightarrow Q) = F$ である。

例： $(\neg P) \rightarrow Q$ の真理値表

P	Q	$\neg P$	$(\neg P) \rightarrow Q$
T	T		
T	F		
F	T		
F	F		

例： $(\neg P) \rightarrow Q$ の真理値表

P	Q	$\neg P$	$(\neg P) \rightarrow Q$
T	T	F	T
T	F	F	T
F	T	T	T
F	F	T	F

例： $(P \wedge Q) \wedge (R \wedge S)$ の真理値表

P	Q	R	S	$P \wedge Q$	$R \wedge S$	$(P \wedge Q) \wedge (R \wedge S)$
T	T	T	T	T	T	T
T	T	T	F	T	F	F
T	T	F	T	T	F	F
T	T	F	F	T	F	F
T	F	T	T	F	T	F
T	F	T	F	F	F	F
T	F	F	T	F	F	F
T	F	F	F	F	F	F
F	T	T	T	F	T	F
F	T	T	F	F	F	F
F	T	F	T	F	F	F
F	T	F	F	F	F	F
F	F	T	T	F	T	F
F	F	T	F	F	F	F
F	F	F	T	F	F	F
F	F	F	F	F	F	F

同値

定義

φ と ψ を論理式とする。任意の真理値割り当て V に対し $V(\varphi) = V(\psi)$ が成り立つとき、 φ と ψ は同値 (*equivalent*) である。

例

$(\neg P) \rightarrow Q$ と $P \vee Q$ が同値である：

P	Q	$\neg P$	$(\neg P) \rightarrow Q$	$P \vee Q$
T	T	F	T	T
T	F	F	T	T
F	T	T	T	T
F	F	T	F	F

演算子の完全性

定義

X を演算子の集合とする。任意の論理式 φ に対し X の演算子のみを用いた論理式 ψ が存在して φ と ψ が同値になるとき、 X を完全 (*functionally complete*) と呼ぶ。

例

- ▶ $\{\neg, \rightarrow\}$ は完全である。
- ▶ $\{\rightarrow\}$ は完全でない。

$\vee$ と $\wedge$ を $\neg$ と $\rightarrow$ で表せる

P	Q	$\neg P$	$(\neg P) \rightarrow Q$	$P \vee Q$
T	T	F	T	T
T	F	F	T	T
F	T	T	T	T
F	F	T	F	F

P	Q	$\neg Q$	$P \rightarrow (\neg Q)$	$\neg(P \rightarrow (\neg Q))$	$P \wedge Q$
T	T	F	F	T	T
T	F	T	T	F	F
F	T	F	T	F	F
F	F	T	T	F	F

$\{\neg, \rightarrow\}$ は完全である

論理式に表す $\vee$ と $\wedge$ を書き直せば、

定理

$\{\neg, \rightarrow\}$ は完全である。

例

$P \vee (Q \wedge R)$ と $(\neg P) \rightarrow (\neg(Q \rightarrow (\neg R)))$ は同値である。

$\{\rightarrow\}$ は完全でない

$P \mid P \rightarrow P$	
T	T
F	T

φ が P と $\rightarrow$ のみを用いる論理式とする。

- ▶ $V(P) = T$ ならば、 $V(\varphi) = T$ である。
- ▶ $V(\varphi) = V(\neg P)$ は成り立たない。

定理

$\{\rightarrow\}$ は完全でない。

まとめ

- ▶ 命題論理の論理式と意味論を定義した。
- ▶ 意味論に基づいて、同値を導入した。
- ▶ 演算子の完全性を調べてみた。

次に演習問題に取り組む。

演習問題 1

問題

演算子 $NAND$ を $|$ と書く。 $NAND$ の真理値表は：

P	Q	$P Q$
T	T	F
T	F	T
F	T	T
F	F	T

演算子の集合 $\{|$ $\}$ が完全であることを証明せよ。

目的：

- ▶ 真理値表の理解の確認。
- ▶ 演算子の完全性の理解の確認。

演習問題 2

問題

演算子 XOR を $\leftrightarrow$ と書く。 XOR の真理値表は：

P	Q	$P \leftrightarrow Q$
T	T	F
T	F	T
F	T	T
F	F	F

演算子の集合 $\{\leftrightarrow\}$ が完全であることを証明せよ。

目的：

- ▶ さらに意味論と演算子の完全性の理解を深める。

ありがとうございます！

演習問題 1 : 解答

P	$P \mid P$	$\neg P$
T	F	F
F	T	T

P	Q	$Q \mid Q$	$P \mid (Q \mid Q)$	$P \rightarrow Q$
T	T	F	T	T
T	F	T	F	F
F	T	F	T	T
F	F	T	T	T

演習問題 2 : 解答

$P \mid P \leftrightarrow P$	
T	F
F	F

φ が P と $\leftrightarrow$ のみを用いる論理式とする。

- ▶ $V(P) = F$ ならば、 $V(\varphi)$ である。
- ▶ $V(\varphi) = V(\neg P)$ は成り立たない。

定理

$\{\leftrightarrow\}$ は完全でない。

REFERENCES I

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